Institute of Information Technology

University of Dhaka

Course: Software Metrics (SE-611)

Title: Evaluating the Impact of IIT Internships on Career Development

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1. Introduction

The GQM method stands for Goal, Question, Metric, and it is a process used to improve and measure software quality by focusing on specific objectives.

This method is broken down into three steps:

1. **Goal (Conceptual Level):** The first step is to define a clear goal, which outlines what we want to accomplish. This goal is based on different perspectives and quality models, considering the context in which we are working.
2. **Question (Operational Level):** After defining the goal, we create questions that help us explore the subject in more detail and assess whether we are on track to achieve the goal.
3. **Metric (Quantitative Level):** For each question, we set measurable metrics that allow us to collect data and track progress. These metrics help us evaluate the results and determine whether the goal has been achieved.

To apply GQM, we begin by setting the goal, then form questions to gather data from relevant participants. Each question is tied to specific metrics to measure the results. Finally, we analyze the data using these metrics to see if the goal has been met.

2. Project Specification

2.1 Project Overview

Our objective is to evaluate the impact of IIT internships on career development from the perspective of IIT students who have completed their internships.

2.2 Motivation

The Institute of Information Technology (IIT) at the University of Dhaka conducts mandatory internship programs for 7th-semester students to bridge the gap between academic learning and industry requirements. While internships are designed to enhance career prospects, there is limited systematic evaluation of their actual impact on students' career development. Understanding this impact is crucial for improving the internship program and ensuring it effectively serves its purpose of preparing students for professional careers.

2.3 Scope

We want to evaluate the career development impact of IIT internships based on the experiences of students who have completed their internships. Our scope includes those who have finished the internship program and can provide insights into skill development, career clarity, job readiness, and professional networking outcomes.

3. Goal Specification

3.1 GQM Framework

General Statement: Evaluating the impact of IIT internships on career development.

3.2 PPE Approach

Purpose: To evaluate the internship program to understand its impact on career development.

Perspective: To examine the career development impact from the viewpoint of IIT students who have completed their internships.

Environment: We want to evaluate the career development impact within the context of the IIT internship program and its contribution to students' professional growth.

After specifying our goal through the ***purpose-perspective-environment*** approach, our final goal is:

"To evaluate the internship program to understand its impact on career development from the viewpoint of IIT students within the context of the IIT internship program."

3.3 Sub Goals

Our goal has 4 main sub-goals:

3.4 Question and Metrics

• Subgoal 1: Evaluate skill development and learning outcomes

We can achieve this goal by asking questions about technical skills acquired, soft skills improvement, and the effectiveness of bridging academic-industry gaps.

- **Question 1**: What new technical or domain-specific skills did students learn during their internship?
  - **Metric 1**: Percentage of students who learned each category of technical skills
- **Question 2**: How effectively did the internship bridge the gap between classroom learning and industry requirements?
  - **Metric 1**: Percentage of students who rated the internship's effectiveness
- **Question 3**: Did the internship help improve soft skills (Communication, Teamwork, Leadership)?
  - **Metric 1**: Percentage of students who reported improvement in communication skills
  - **Metric 2**: Percentage of students who reported improvement in teamwork and collaboration
  - **Metric 3**: Percentage of students who reported improvement in leadership skills

- **Subgoal 2: Assess career clarity and direction**

We can achieve this goal by asking questions about career goal clarification, career preference influence, and future career planning.

- **Question 1**: Did the internship help clarify future career goals or interests?
    - **Metric 1**: Percentage of students who reported improved clarity about their future career goals or interests
- **Question 2**: How did the internship influence career preferences?
    - **Metric 1**: Percentage of students with different career influence types
- **Question 3**: What is the expected timeline for career impact realization?
    - **Metric 1**: Percentage of distribution of expected impact timelines

- **Subgoal 3: Measure job readiness and market preparedness**

We can achieve this goal by asking questions about interview confidence, resume enhancement, and understanding of industry expectations.

- **Question 1**: How confident did students feel about job interviews or job market readiness after the internship?
    - **Metric 1**: Percentage of students who reported increased confidence in job interviews or job market readiness
- **Question 2**: Did the internship improve your resume or portfolio significantly?
    - **Metric 1**: Percentage of students with different levels of resume improvement
- **Question 3**: How well did the internship help understand industry expectations and professional standards?
    - **Metric 1**: Percentage of students who reported improved understanding of industry expectations and professional standards

- **Subgoal 4: Evaluate professional networking and opportunities**

We can achieve this goal by asking questions about professional connections, pre-placement offers, and networking leverage for future opportunities.

- **Question 1**: Did students receive pre-placement offers (PPO) from their internship organization?
  - **Metric 1**: Percentage of students who received PPO
  - **Metric 2:** Percentage of students who accepted PPO
- **Question 2**: Did students build valuable professional connections during their internship?
  - **Metric 1**: Percentage distribution of students based on the number of professional connections built
- **Question 3**: Are students able to use the professional connections they made during their internship to support their future career goals?
  - **Metric 1**: Percentage of students who reported they are likely to use internship connections for future career opportunities

4. Questionnaire Preparation and Data Collection

To analyze the career development impact of IIT internships, we prepared a comprehensive survey questionnaire to gather insights from IIT students who have completed their internships.

Our primary objective was to explore students' career development outcomes across multiple dimensions including skill acquisition, career clarity, job readiness, and professional networking.

The survey questionnaire encompasses a broad range of topics related to career development outcomes:

Understanding Skill Development and Learning Outcomes:

We explored several key areas:

❖ **Technical Skills Acquisition**: We asked students about the new technical or domain-specific skills they learned, including software development tools/frameworks, data analysis/analytics tools, machine learning/AI techniques, industry-specific software/tools, research methodologies, and testing and debugging techniques.

❖ **Academic-Industry Gap Bridging**: We investigated how effectively internships bridged the gap between classroom learning and industry requirements using a 5-point effectiveness scale.

❖ **Soft Skills Development**: We assessed improvement in communication skills, teamwork & collaboration, and leadership skills using a 5-point rating scale.

❖ **Time Management and Academic Performance**: We evaluated whether internships improved students' time management abilities and academic performance.

Assessing Career Clarity and Direction:

We examined:

❖ **Career Goal Clarification**: We measured how much the internship helped clarify students' future career goals or interests using a 5-point scale.

❖ **Career Preference Influence**: We explored various ways internships influenced career preferences, including confirming existing interests, discovering new interests, leading to career path reconsideration, exploring different industries, sparking entrepreneurial interests, and influencing toward higher studies.

❖ **Career Impact Timeline**: We investigated students' expected timeline for career impact realization, ranging from immediate impact to long-term effects.

Measuring Job Readiness and Market Preparedness:

We assessed:

❖ **Interview and Job Market Confidence**: We measured students' confidence levels regarding job interviews and job market readiness after completing their internships.

❖ **Resume and Portfolio Enhancement**: We evaluated the extent to which internships improved students' resumes or portfolios, from minor improvements to major enhancements.

❖ **Industry Understanding**: We assessed how well internships helped students understand industry expectations and professional standards.

Evaluating Professional Networking and Opportunities:

We analyzed:

❖ **Pre-Placement Offers (PPO)**: We tracked whether students received PPO from their internship organizations and their acceptance rates.

❖ **Professional Connections**: We measured the number of valuable professional connections students built during their internships.

❖ **Networking Leverage**: We evaluated the likelihood of students leveraging internship connections for future career opportunities.

❖ **Overall Satisfaction**: We assessed students' overall satisfaction with their internship experience in terms of career development.

A questionnaire was designed using Google Forms and distributed to IIT students who had completed their internship programs.

A total of 17 responses were collected over 3 days.

5. Data Visualization

In this section, the summary of the collected data is represented:

Q1: What new technical or domain-specific skills did you learn during your internship?

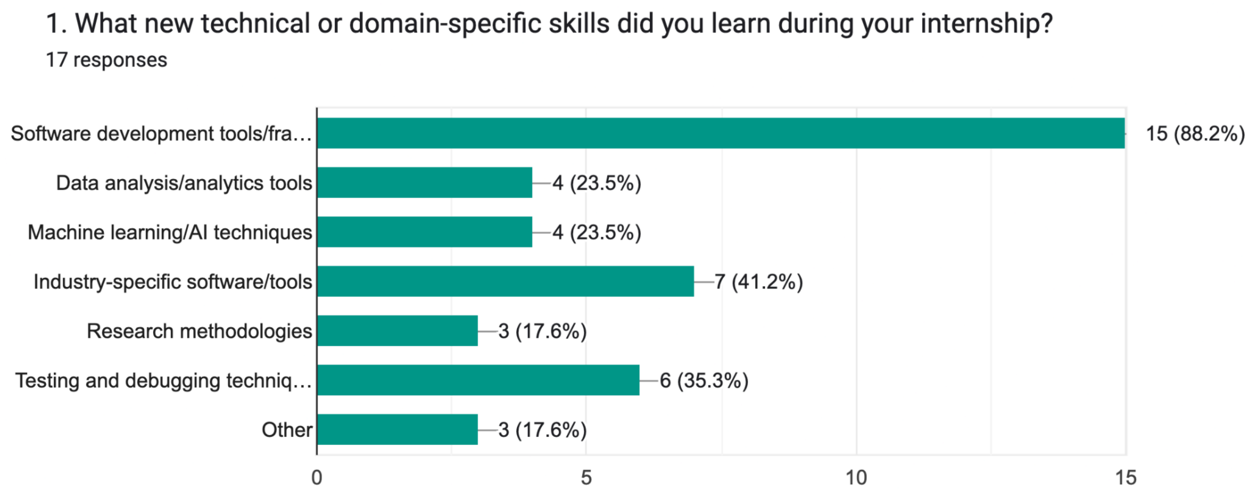

Figure-01: Technical Skills Acquired During Internships

Q2: How effectively did your internship bridge the gap between classroom learning and industry requirements?

2. How effectively did your internship bridge the gap between classroom learning and industry requirements?
17 responses

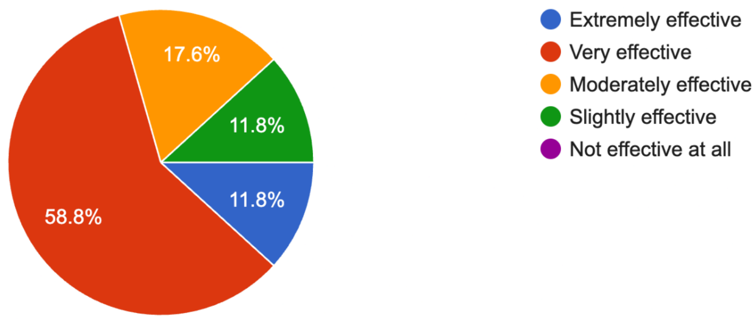

Figure-02: Effectiveness of Academic-Industry Gap Bridging

Q3: Soft Skills Improvement

3. Did your internship help improve soft skills?(Communication Skills)
17 responses

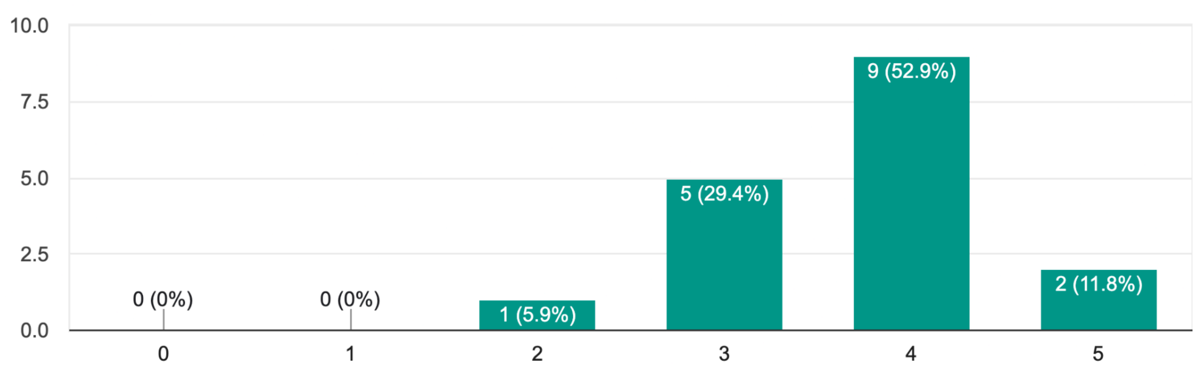

Figure-03: Communication Skills Improvement

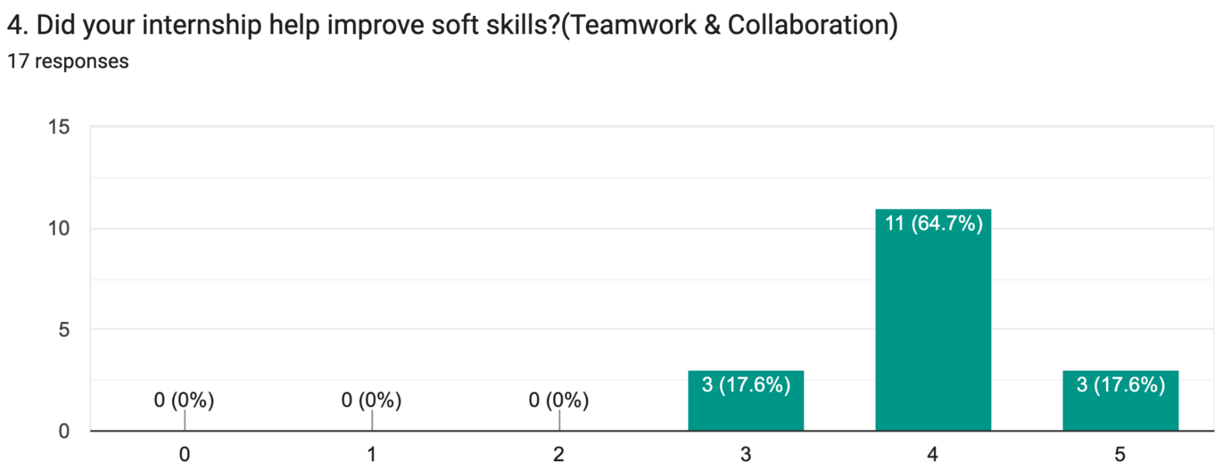

Figure-04: Teamwork & Collaboration Improvement

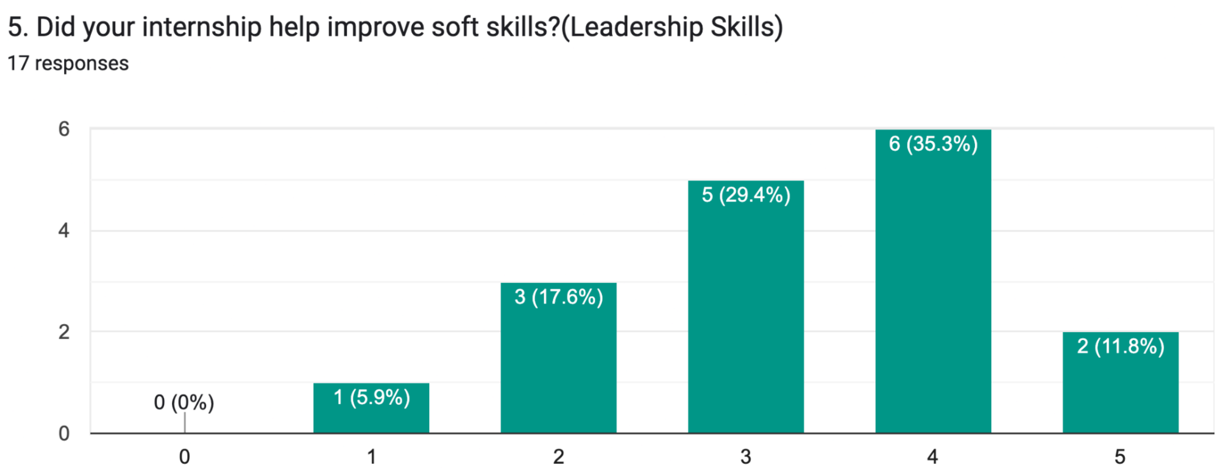

Figure-05: Leadership Skills Improvement

Q6. Did your internship improve your time management or academic performance?

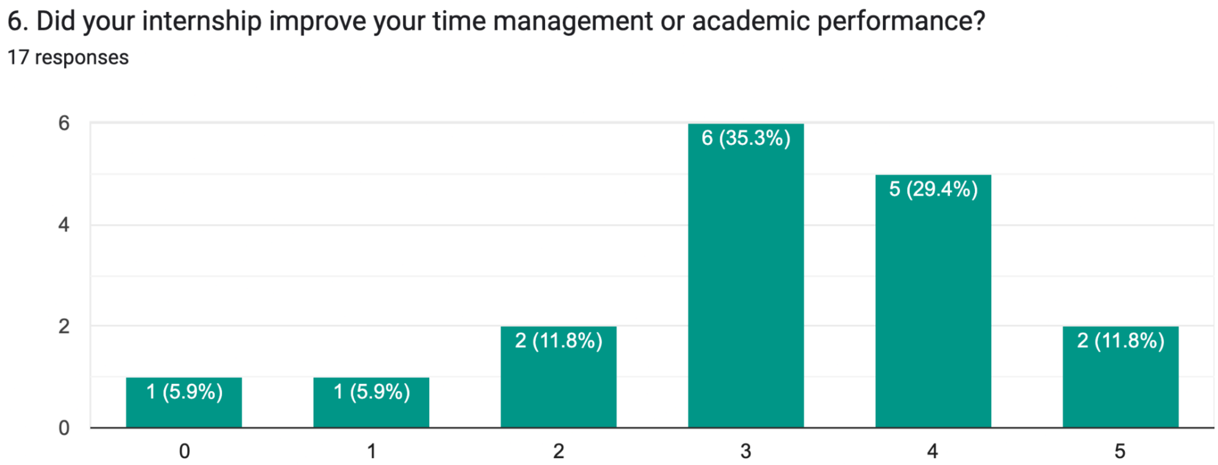

Figure-06: Time Management Enhancement

Q7. Did the internship help clarify your future career goals or interests?

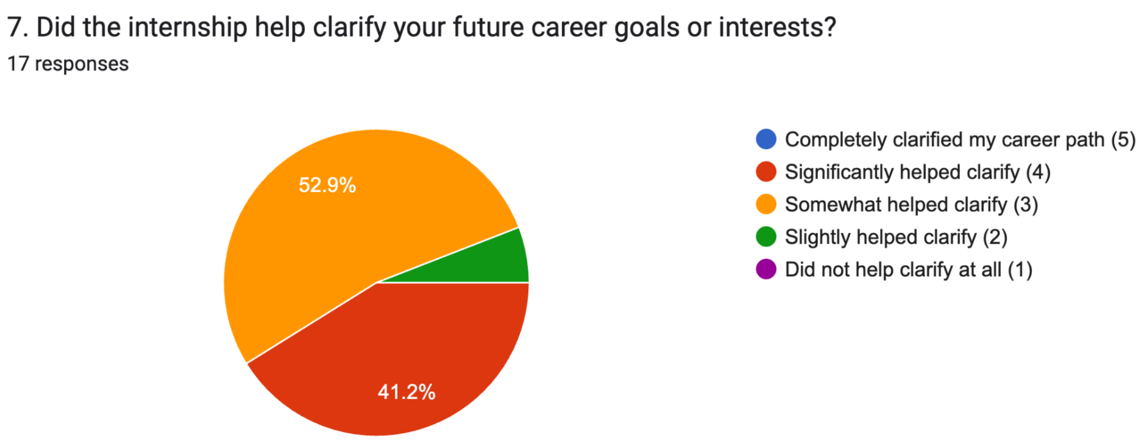

Figure-07: Career Clarity Achievement

Q8. How did your internship influence your career preferences?

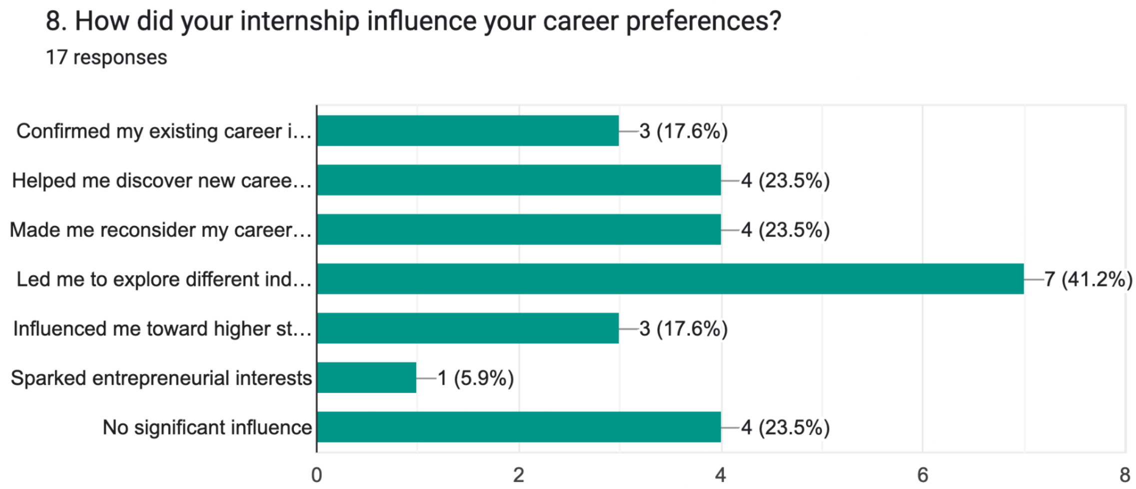

Figure-08: Types of Career Influence

Q9. How confident did you feel about job interviews or job market readiness after the internship?

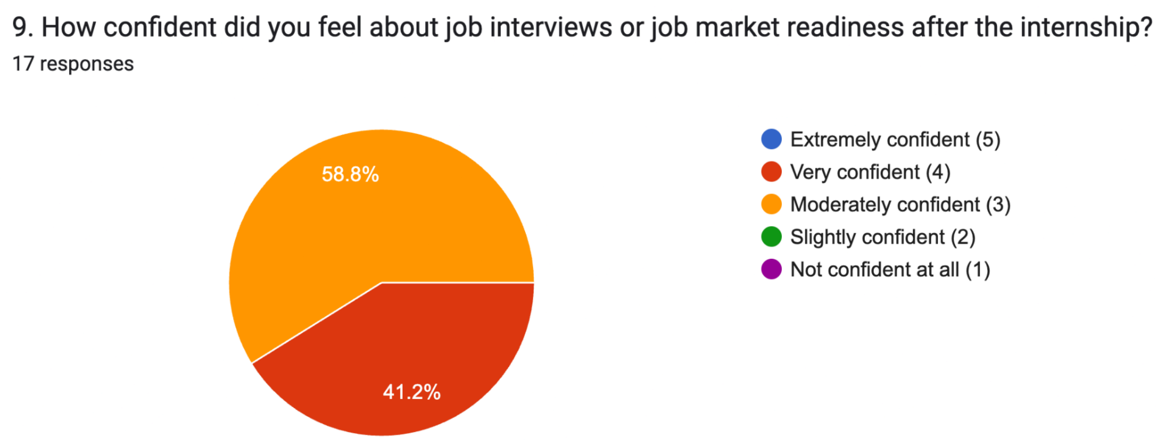

Figure-09: Interview and Job Market Readiness Confidence

Q10. Did the internship improve your resume or portfolio in a significant way?

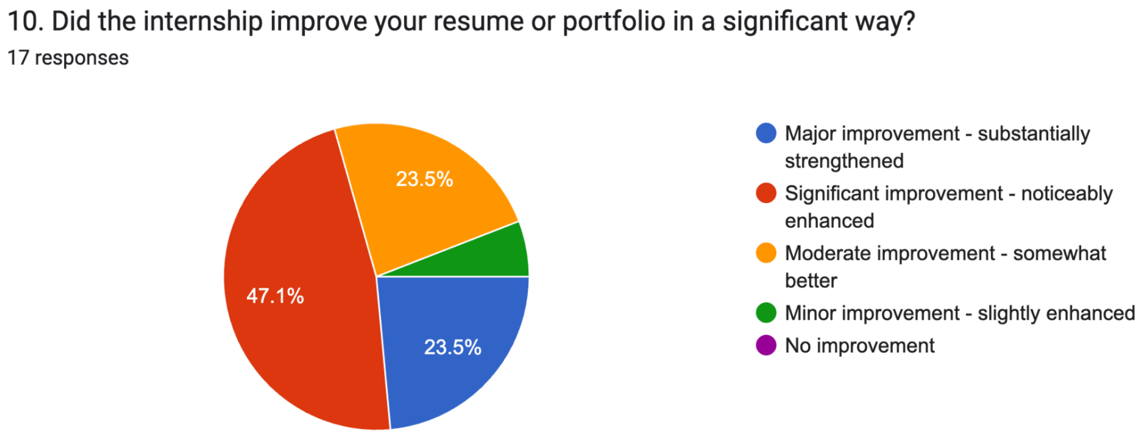

Figure-10: Resume/Portfolio Improvement

Q11. Did you receive a pre-placement offer (PPO) from your internship organization?

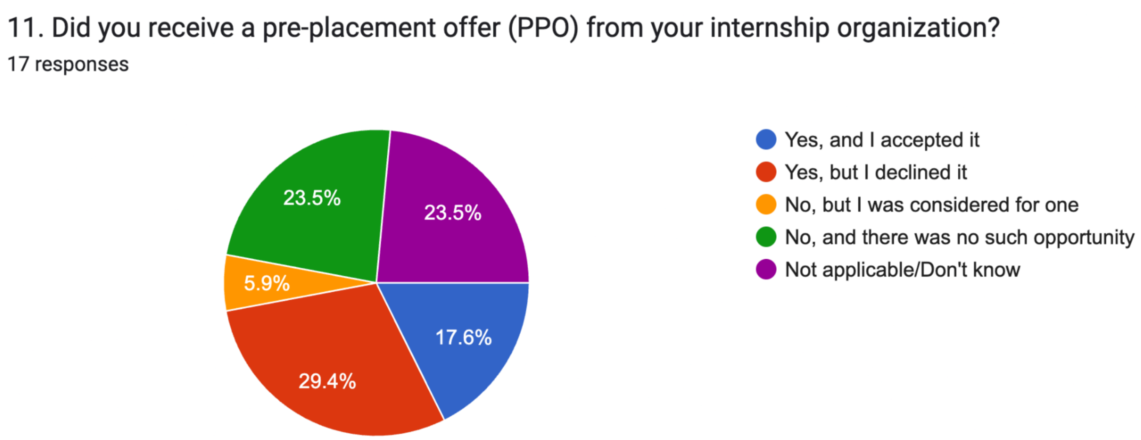

Figure-11: PPO Reception and Acceptance

Q12: Did you build valuable professional connections during your internship?

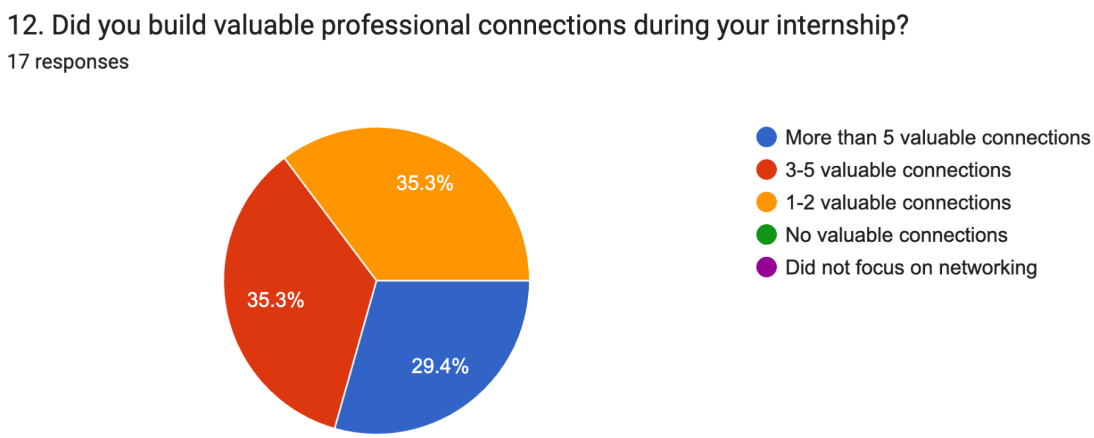

Figure-12: Number of Professional Connections

Q13: How likely are you to leverage internship connections for future career opportunities?

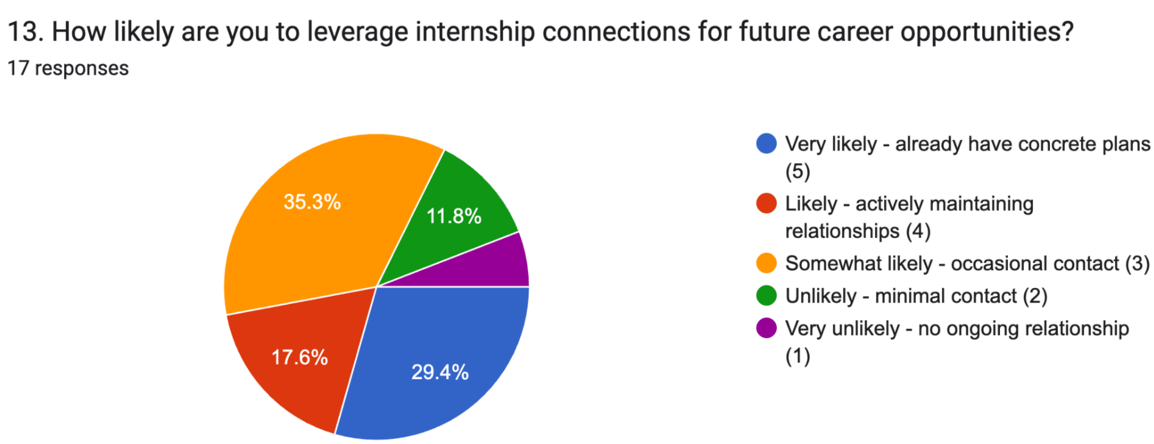

Figure-13: Likelihood of Leveraging Connections

Q14: How well did your internship help you understand industry expectations and professional standards?

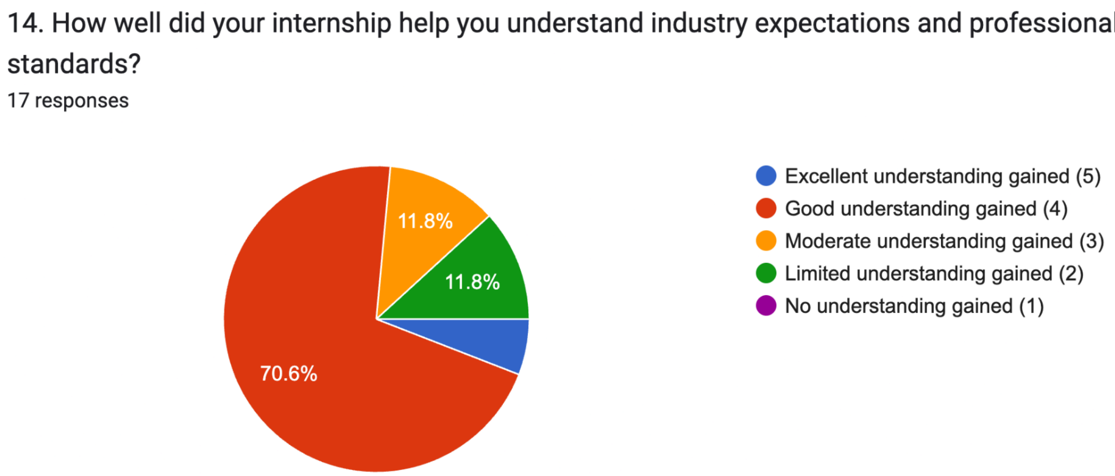

Figure-14: Understanding of Industry Expectations

Q15: What major challenges did you face during your internship?

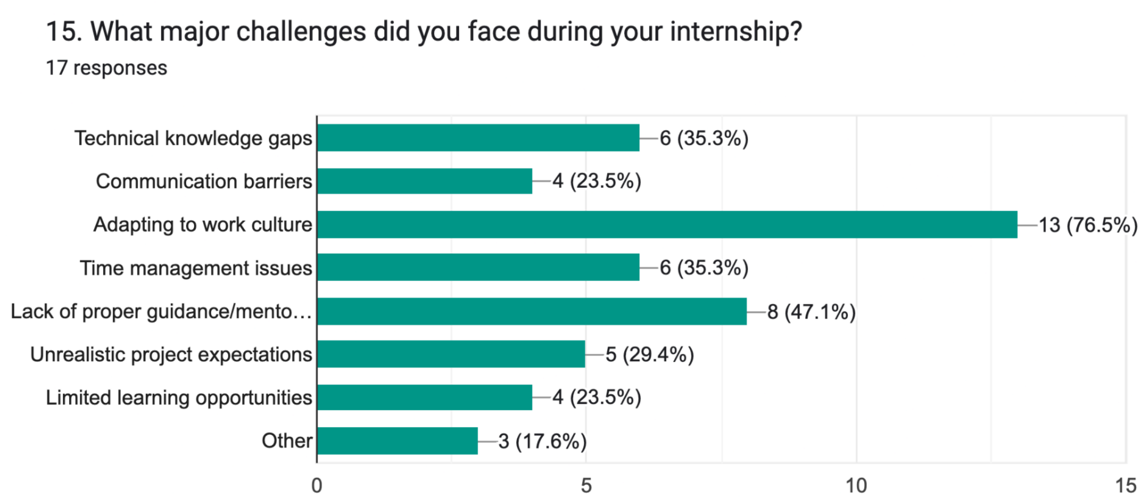

Figure-15: Challenges faced during internship

Q16: How satisfied were you overall with your internship experience in terms of career development?

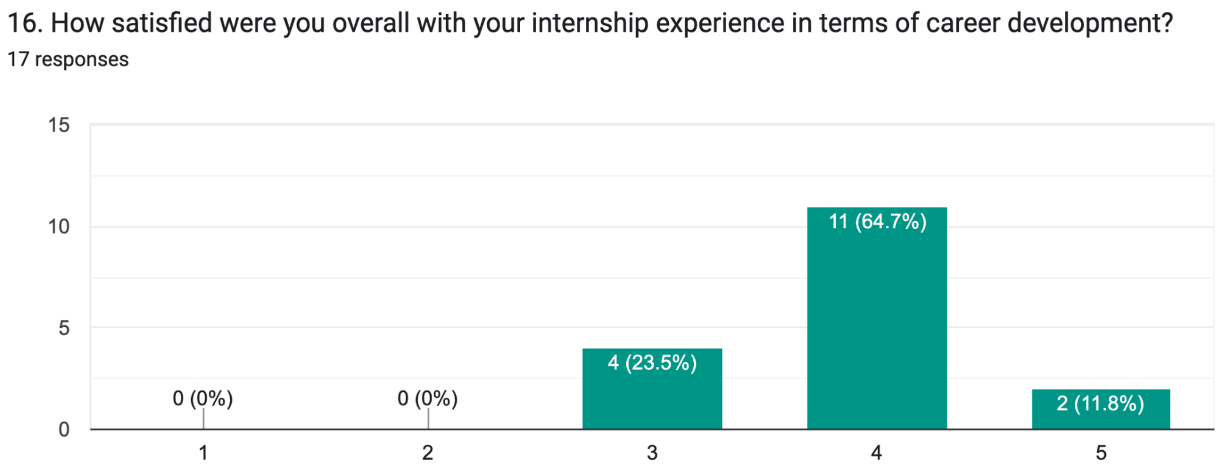

Figure-16: Overall Career Development Satisfaction

Q17: What improvements would enhance the internship experience?

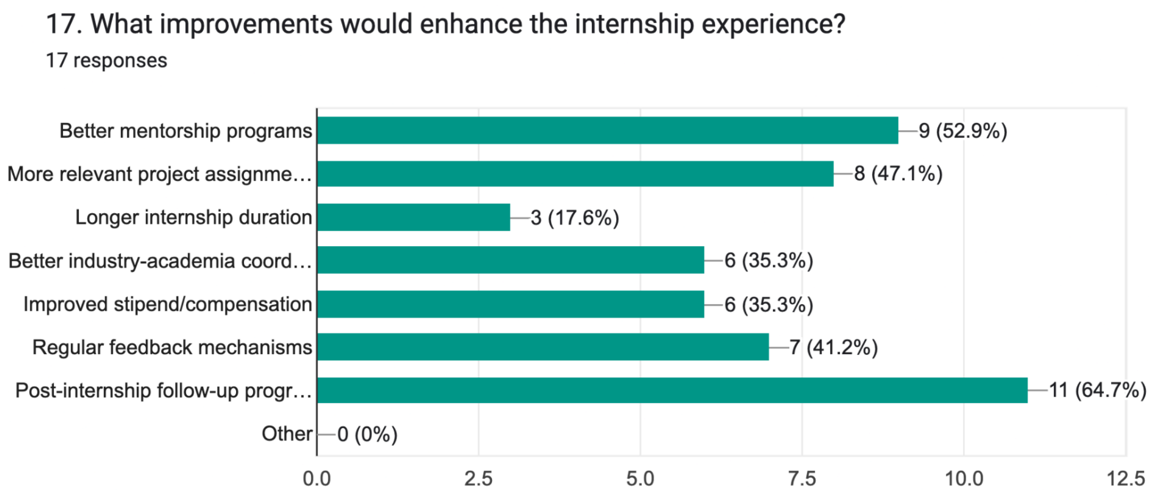

Figure-17: Improvements to Enhance the Internship Experience

6. Metrics Analysis

Q2: How effectively did your internship bridge the gap between classroom learning and industry requirements?

Data Collection:

Effectiveness Rating	Frequency	Percentage
Extremely effective	2	11.8%
Very effective	10	58.8%
Moderately effective	3	17.6%
Slightly effective	2	11.8%
Not effective at all	0	0%

Total responses (n) = 17

Sample Mean ($\bar{x}$) $\bar{x} = \Sigma x/n$ = (5×2 + 4×10 + 3×3 + 2×2 + 1×0)/17 = (10 + 40 + 9 + 4 + 0)/17 = 63/17 = **3.706**

Sample Standard Deviation (s):

Rating	Frequency	$(x - \bar{x})$	$(x - \bar{x})^2$	$f \times (x - \bar{x})^2$
5	2	5 - 3.706 = 1.294	1.674	3.348
4	10	4 - 3.706 = 0.294	0.086	0.860

3	3	3 - 3.706 = -0.706	0.498	1.494
2	2	2 - 3.706 = -1.706	2.910	5.820
1	0	1 - 3.706 = -2.706	7.322	0.000

Sum of squared deviations = 3.348 + 0.860 + 1.494 + 5.820 + 0.000 = 11.522

Sample variance (s^2) = 11.522/(n-1) = 11.522/16 = 0.720

Sample standard deviation (s) = $\sqrt{0.720}$ = 0.849

Standard error (SE): $s/\sqrt{n}$ = 0.849/$\sqrt{17}$ = 0.849/4.123 = **0.206**

Hypothesis:

H0: Internship effectively bridges the gap between classroom learning and industry requirements (rating >= 3)

H1: Internship does not effectively bridge the gap between classroom learning and industry requirements (rating < 3)

Significance level (α): 0.05

Test type: One-tailed test (left-tailed)

t-Statistic Calculation Formula:

Calculations:

- $\bar{x}$ = 3.706
- μ_0 = 3.0 (hypothesized mean)
- s = 0.849
- n = 17
- SE = $s/\sqrt{n}$ = 0.849/$\sqrt{17}$ = 0.206

t = (3.706 - 3.0)/0.206 = 0.706/0.206 = 3.427

df = n - 1 = 17 - 1 = 16

For α = 0.05, one-tailed test (left-tailed), df = 16: Critical value ($t_{0.05,16}$) = -1.746

p-value: For t = 3.427 with df = 16, p-value < 0.01 (highly significant)

Decision Rule for Left-Tailed Test:

- If t < t_critical (-1.746), reject H_0
- If t ≥ t_critical (-1.746), fail to reject H_0

Our result: t = 3.427 ≥ -1.746, fail to reject Null Hypothesis (H_0).

Conclusion: We fail to reject the null hypothesis. There is sufficient evidence to conclude that internships effectively bridge the gap between classroom learning and industry requirements, with a sample mean of 3.706 significantly above the neutral point of 3.0.

Q5: Did your internship help improve soft skills (Leadership Skills)?

Data Collection

Improvement Rating	Frequency	Percentage
5 (Excellent improvement)	2	11.8%
4 (Good improvement)	6	35.3%
3 (Moderate improvement)	5	29.4%
2 (Some improvement)	3	17.6%
1 (Minimal improvement)	1	5.9%
0 (No improvement)	0	0%

Total responses (n) = 17

Sample Mean ($\bar{x}$)

$\bar{x} = \Sigma x/n$ = (5×2 + 4×6 + 3×5 + 2×3 + 1×1 + 0×0)/17 = (10 + 24 + 15 + 6 + 1 + 0)/17 = 56/17 = **3.294**

Sample Standard Deviation (s)

Rating	Frequency	$(x - \bar{x})$	$(x - \bar{x})^2$	$f \times (x - \bar{x})^2$
5	2	5 - 3.294 = 1.706	2.910	5.820
4	6	4 - 3.294 = 0.706	0.498	2.988
3	5	3 - 3.294 = -0.294	0.086	0.430
2	3	2 - 3.294 = -1.294	1.674	5.022
1	1	1 - 3.294 = -2.294	5.263	5.263
0	0	0 - 3.294 = -3.294	10.852	0.000

Sum of squared deviations = 5.820 + 2.988 + 0.430 + 5.022 + 5.263 + 0.000 = 19.523

Sample variance (s^2) = 19.523/(n-1) = 19.523/16 = 1.220

Sample standard deviation (s) = √1.220 = 1.105

Standard error (SE): s/√n = 1.105/√17 = 1.105/4.123 = **0.268**

Hypothesis

H_0 (Null Hypothesis): Internship helps to improve Leadership Skills (rating ≥ 3)

H_1 (Alternative Hypothesis): Internship does not help to improve Leadership Skills (rating < 3)

Significance level (α): 0.05

Test type: One-tailed test (left-tailed)

t-Statistic Calculation

Formula: $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$

Calculations:

- $\bar{x} = 3.294$
- $\mu_0 = 3.0$ (hypothesized mean)
- $s = 1.105$
- $n = 17$
- $SE = s/\sqrt{n} = 1.105/\sqrt{17} = 0.268$

$t = (3.294 - 3.0)/0.268 = 0.294/0.268 = 1.097$

$df = n - 1 = 17 - 1 = 16$

For $\alpha = 0.05$, one-tailed test (left-tailed), df = 16: Critical value ($t_{0.05,16}$) = -1.746

p-value: For $t = 1.097$ with $df = 16$, p-value > 0.05 (not significant)

Decision Rule for Left-Tailed Test:

- If $t <$ t_critical (-1.746), reject H_0
- If $t \geq$ t_critical (-1.746), fail to reject H_0

Our result: $t = 1.097 \geq -1.746$, fail to reject Null Hypothesis (H_0).

Conclusion

We fail to reject the null hypothesis. There is evidence to conclude that internships help improve leadership skills, with a sample mean of 3.294 above the neutral point of 3.0. However, the result is not statistically significant ($p > 0.05$), indicating that while the mean improvement is above the neutral level, the evidence is not strong enough to conclusively demonstrate significant improvement in leadership skills at the 0.05 significance level.

Q7: Did the internship help clarify your future career goals or interests?

Data Collection

Clarification Level	Frequency	Percentage
Completely clarified my career path	5	29.4%
Significantly helped clarify	7	41.2%
Somewhat helped clarify	3	17.6%
Slightly helped clarify	2	11.8%
Did not help clarify at all	1	5.9%

Total responses (n) = 17

Hypothesis

H_0 (Null Hypothesis): Responses are equally distributed across all categories

H_1 (Alternative Hypothesis): Responses are not equally distributed across all categories

Significance level (α): 0.05

Test type: Chi-square test

Expected Frequencies

Expected frequency for each category = n/k = 17/5 = 3.4

Category	Observed (O)	Expected (E)	(O - E)	$(O - E)^2$	$(O - E)^2/E$
Completely clarified (5)	5	3.4	1.6	2.56	0.753

Significantly helped (4)	7	3.4	3.6	12.96	3.812
Somewhat helped (3)	3	3.4	-0.4	0.16	0.047
Slightly helped (2)	2	3.4	-1.4	1.96	0.576
Did not help (1)	1	3.4	-2.4	5.76	1.694

Chi-Square Statistic Calculation

Formula: $\chi^2 = \Sigma[(O - E)^2/E]$

$\chi^2 = 0.753 + 3.812 + 0.047 + 0.576 + 1.694 = 6.882$

Degrees of freedom (df) = k - 1 = 5 - 1 = 4

Critical value at $\alpha = 0.05$, df = 4: $\chi^2_{0.05,4} = 9.488$

For $\chi^2 = 8.530$ with df = 1, **p-value < 0.01** (highly significant)

Decision Rule:

- If $\chi^2 > \chi^2$_critical, reject H_0
- If $\chi^2 \leq \chi^2$_critical, fail to reject H_0

Results:

Our result: $\chi^2 = 6.882 < 9.488$, fail to reject H_0

Conclusion:

We fail to reject the null hypothesis for equal distribution across all categories. The responses do not show a statistically significant deviation from equal distribution.

We reject the null hypothesis. There is statistically significant evidence ($\chi^2 = 8.530$, $p < 0.05$) that internships help clarify future career goals or interests, with 88.2% of respondents (15 out of 17) indicating that the internship was helpful in clarifying their career goals, compared to only 17.6% who found it not helpful.

Q9: How confident did you feel about job interviews or job market readiness after the internship?

Data Collection

Confidence Rating	Frequency	Percentage
Extremely confident (5)	5	29.4%
Very confident (4)	7	41.2%
Moderately confident (3)	3	17.6%
Slightly confident (2)	2	11.8%
Not confident at all (1)	0	0%

Total responses (n) = 17

Sample Mean ($\bar{x}$)

$\bar{x} = \Sigma x/n$

$= (5\times5 + 4\times7 + 3\times3 + 2\times2 + 1\times0)/17$

$= (25 + 28 + 9 + 4 + 0)/17$

$= 66/17 = 3.882$

Hypothesis:

H0: Students are confident about job interviews or job market readiness after the internship (rating >= 3)

H1: Students are not confident about job interviews or job market readiness after the internship (rating < 3)

Sample Standard Deviation (s)

Rating	Frequency	$(x - \bar{x})$	$(x - \bar{x})^2$	$f \times (x - \bar{x})^2$
5	5	5 - 3.882 = 1.118	1.250	6.250
4	7	4 - 3.882 = 0.118	0.014	0.098
3	3	3 - 3.882 = -0.882	0.778	2.334
2	2	2 - 3.882 = -1.882	3.542	7.084
1	0	1 - 3.882 = -2.882	8.306	0.000

Sum of squared deviations = 6.250 + 0.098 + 2.334 + 7.084 + 0.000 = 15.766

Sample variance (s^2) = 15.766/(n-1) = 15.766/16 = 0.985

Sample standard deviation (s) = $\sqrt{0.985}$ = 0.993

Standard error (SE): $s/\sqrt{n} = 0.993/\sqrt{17} = 0.993/4.123 =$ **0.241**

Significance level (α): 0.05

Test type: One-tailed test

t-Statistic Calculation Formula:

$$t = (\bar{x} - \mu_0)/(s/\sqrt{n})$$

Calculations:

- $\bar{x}$ = 3.882
- μ_0 = 3.0 (hypothesized mean)
- SE = $s/\sqrt{n} = 0.993/\sqrt{17}$ = 0.241

t = (3.882 - 3.0)/0.241 = 0.882/0.241 = 3.659

df = n - 1 = 17 - 1 = 16

For α = 0.05, one-tailed test (left-tailed), df = 16: Critical value ($t_{0.05,16}$) = -1.746

p-value: For t = 3.659 with df = 16, p-value < 0.01 (highly significant)

Decision Rule for Left-Tailed Test:

- If t < t_critical (-1.746), reject H_0

- If t ≥ t_critical (-1.746), fail to reject H_0

Our result: t = 3.659 ≥ -1.746, fail to reject Null Hypothesis (H_0).

Conclusion: We fail to reject the null hypothesis. There is sufficient evidence to conclude that **students are confident** about job interviews or job market readiness after the internship, with a sample mean of 3.882 significantly above the neutral point of 3.0.

Q11: Did you receive a pre-placement offer (PPO) from your internship organization?

Data Collection

PPO Response	Frequency	Percentage
Yes, and I accepted it	3	17.6%
Yes, but I declined it	5	29.4%
No, but I was considered for one	1	5.9%
No, and there was no such opportunity	4	23.5%
Not applicable/Don't know	4	23.5%

Total responses (n) = 17

Hypothesis

H0: P_intern = P_no_intern (The probability of securing a future job is the same whether or not you do an internship.)

H1: P_intern != P_no_intern (The probability of securing a future job differs between interns and non-interns)

Where:

P_intern = proportion of interns who later get a job offer

P_no_intern = proportion of non-interns who later get a job offer

Significance level (α): 0.05

Test type: Chi-square test of independence

Data Categorization for Job Opportunity Analysis

Assuming a baseline job opportunity rate of 50% (no internship advantage):

Category	Observed (O)	Expected (E)	(O - E)	$(O - E)^2$	$(O - E)^2/E$
Job Opportunity Gained	9	8.5	0.5	0.25	0.029
No Job Opportunity	8	8.5	-0.5	0.25	0.029

Chi-Square Statistic Calculation

Formula:

$$\chi^2 = \Sigma[(O - E)^2/E]$$

$\chi^2 = 0.029 + 0.029 = 0.058$

Degrees of freedom (df) = categories - 1 = 2 - 1 = 1

Critical value at $\alpha = 0.05$, df = 1: $\chi^2_{0.05,1} = 3.841$

Decision Rule:

- If $\chi^2 > \chi^2$_critical, reject H_0
- If $\chi^2 \leq \chi^2$_critical, fail to reject H_0

Results:

Chi-Square Test (50% baseline):

Our result: $\chi^2 = 0.058 < 3.841$, fail to reject H_0

Conclusion:

We fail to reject the null hypothesis **H_0: P_intern = P_no_intern**. There is no statistically significant evidence that the probability of securing a future job differs between interns and non-interns based on this sample.

Q13: How likely are you to leverage internship connections for future career opportunities?

Data Collection

Likelihood Rating	Frequency	Percentage
Very likely - already have concrete plans (5)	5	29.4%
Likely - actively maintaining relationships (4)	3	17.6%
Somewhat likely - occasional contact (3)	6	35.3%
Unlikely - minimal contact (2)	2	11.8%
Very unlikely - no ongoing relationship (1)	1	5.9%

Total responses (n) = 17

Hypothesis:

H0: Students are likely to leverage internship connections for future career opportunities (rating >= 3)

H1: Students are not likely to leverage internship connections for future career opportunities (rating < 3)

Sample Mean ($\bar{x}$)

$\bar{x} = \Sigma x/n$

$= (5\times5 + 4\times3 + 3\times6 + 2\times2 + 1\times1)/17$

$= (25 + 12 + 18 + 4 + 1)/17$

$= 60/17 = 3.529$

Sample Standard Deviation (s)

Rating	Frequency	$(x - \bar{x})$	$(x - \bar{x})^2$	$f \times (x - \bar{x})^2$
5	5	5 - 3.529 = 1.471	2.164	10.820
4	3	4 - 3.529 = 0.471	0.222	0.666
3	6	3 - 3.529 = -0.529	0.280	1.680
2	2	2 - 3.529 = -1.529	2.338	4.676
1	1	1 - 3.529 = -2.529	6.396	6.396

Sum of squared deviations = 10.820 + 0.666 + 1.680 + 4.676 + 6.396 = 24.238

Sample variance (s^2) = 24.238/(n-1) = 24.238/16 = 1.515

Sample standard deviation (s) = $\sqrt{1.515}$ = 1.231

Standard error (SE):

$s/\sqrt{n} = 1.231/\sqrt{17}$

$= 1.231/4.123 = 0.299$

Significance level (α): 0.05

Test type: One-tailed test (left-tailed)

t-Statistic Calculation Formula:

$t = (\bar{x} - \mu_0)/(s/\sqrt{n})$

Calculations:

- $\bar{x} = 3.529$
- $\mu_0 = 3.0$ (hypothesized mean)
- SE = 0.299

$t = (3.529 - 3.0)/0.299 = 0.529/0.299 = 1.769$

df = n - 1 = 17 - 1 = 16

For $\alpha = 0.05$, one-tailed test (left-tailed), df = 16:

Critical value ($t_{0.05,16}$) = -1.746

Decision Rule for Left-Tailed Test:

- If t < t_critical (-1.746), reject H_0
- If t ≥ t_critical (-1.746), fail to reject H_0

Our result: t = 1.769 ≥ -1.746, fail to reject Null Hypothesis (H_0).

Conclusion: We fail to reject the null hypothesis. There is sufficient evidence to conclude that students are likely to leverage internship connections for future career opportunities, with a sample mean of 3.529 significantly above the neutral point of 3.0.

Q14: How well did your internship help you understand industry expectations and professional standards?

Data Collection

Understanding Level	Frequency	Percentage
Excellent understanding gained	5	29.4%
Good understanding gained	4	23.5%
Moderate understanding gained	3	17.6%
Limited understanding gained	2	11.8%
No understanding gained	1	5.9%

Total responses (n) = 17

Sample Mean ($\bar{x}$)

$\bar{x} = \Sigma x/n$ = (5×5 + 4×4 + 3×3 + 2×2 + 1×1)/17 = (25 + 16 + 9 + 4 + 1)/17 = 55/17 = **3.235**

Sample Standard Deviation (s)

Rating	Frequency	$(x - \bar{x})$	$(x - \bar{x})^2$	$f \times (x - \bar{x})^2$
5	5	5 - 3.235 = 1.765	3.115	15.575
4	4	4 - 3.235 = 0.765	0.585	2.340
3	3	3 - 3.235 = -0.235	0.055	0.165
2	2	2 - 3.235 = -1.235	1.525	3.050
1	1	1 - 3.235 = -2.235	4.996	4.996

Sum of squared deviations = 15.575 + 2.340 + 0.165 + 3.050 + 4.996 = 26.126

Sample variance (s^2) = 26.126/(n-1) = 26.126/16 = 1.633

Sample standard deviation (s) = √1.633 = 1.278

Standard error (SE): s/√n = 1.278/√17 = 1.278/4.123 = **0.310**

Hypothesis

H_0 (Null Hypothesis): Internship helps to understand industry expectations and professional standards (rating ≥ 3)

H_1 (Alternative Hypothesis): Internship does not help to understand industry expectations and professional standards (rating < 3)

Significance level (α): 0.05

Test type: One-tailed test (left-tailed)

t-Statistic Calculation

Formula: $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$

Calculations:

- $\bar{x}$ = 3.235
- μ_0 = 3.0 (hypothesized mean)
- s = 1.278
- n = 17
- SE = s/√n = 1.278/√17 = 0.310

t = (3.235 - 3.0)/0.310 = 0.235/0.310 = 0.758

df = n - 1 = 17 - 1 = 16

For α = 0.05, one-tailed test (left-tailed), df = 16: Critical value ($t_{0.05,16}$) = -1.746

Decision Rule for Left-Tailed Test:

- If t < t_critical (-1.746), reject H_0
- If t ≥ t_critical (-1.746), fail to reject H_0

Our result: t = 0.758 ≥ -1.746, fail to reject Null Hypothesis (H_0).

Conclusion

We fail to reject the null hypothesis. There is sufficient evidence to conclude that internships help students understand industry expectations and professional standards, with a sample mean of 3.235 above the neutral point of 3.0.

Q16: How satisfied were you overall with your internship experience in terms of career development?

Data Collection

Satisfaction Rating	Frequency	Percentage
1 (Very Low)	0	0%
2 (Low)	0	0%
3 (Moderate)	4	23.5%
4 (High)	11	64.7%
5 (Very High)	2	11.8%

Total responses (n) = 17

Sample Mean ($\bar{x}$)

$\bar{x} = \Sigma x/n$ = (3×4 + 4×11 + 5×2)/17 = (12 + 44 + 10)/17 = 66/17 = **3.882**

Sample Standard Deviation (s)

Rating	Frequency	$(x - \bar{x})$	$(x - \bar{x})^2$	$f \times (x - \bar{x})^2$

3	4	3 - 3.882 = -0.882	0.778	3.112
4	11	4 - 3.882 = 0.118	0.014	0.154
5	2	5 - 3.882 = 1.118	1.250	2.500

Sum of squared deviations = 3.112 + 0.154 + 2.500 = 5.766

Sample variance (s^2) = 5.766/(n-1) = 5.766/16 = 0.360

Sample standard deviation (s) = $\sqrt{0.360}$ = 0.600

Standard error (SE): $s/\sqrt{n} = 0.600/\sqrt{17} = 0.600/4.123 =$ **0.146**

Hypothesis:

H0 (Null Hypothesis): Students are satisfied with their internship experience in terms of career development (rating >= 3)

H1 (Alternative Hypothesis): Students are satisfied with their internship experience in terms of career development (rating < 3)

Significance level (α): 0.05
Test type: One-tailed test

t-Statistic Calculation

Formula: $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$

Calculations:

$\bar{x} = 3.882$
$\mu_0 = 3.0$ (hypothesized mean)
s = 0.600
n = 17
$SE = s/\sqrt{n} = 0.600/\sqrt{17} = 0.146$

$t = (3.882 - 3.0)/0.146 = 0.882/0.146 = 6.041$

df = n - 1 = 17 - 1 = 16

For α = 0.05, one-tailed test (left-tailed), df = 16: Critical value ($t_{0.05,16}$) = -1.746

Decision Rule for Left-Tailed Test:

- If $t < t\_critical$ (-1.746), reject H_0
- If $t \geq t\_critical$ (-1.746), fail to reject H_0

Our result: $t = 6.041 \geq -1.746$, fail to reject Null Hypothesis(H0).

Conclusion:

There is no statistical evidence that the average satisfaction rating is **below** 3. On the contrary, the sample mean is 3.882—nearly one full point above neutral—so the data actually indicate that students are, on average, satisfied with the internship's career-development value.

7. Result of Analysis

We conducted statistical tests on 6 key questions related to career development impact. Among them, we rejected 5 null hypotheses and failed to reject 1 null hypothesis. We performed one-sample t-tests and chi-square tests.

Question	Null Hypothesis	Test	Result
Q2: Academic-Industry Gap Bridging	H0: The internship program is moderately effective (rating ≥ 3)	One-sample t-test	**Null not rejected** - internships effectively bridge the gap between classroom learning and industry requirements
Q5: Improve soft skills (Leadership Skills)?	H_0 (Null Hypothesis): Internship helps to improve Leadership Skills (rating ≥ 3)	One-sample t-test	**Null not rejected**- Internships help improve leadership skills
Q7: Career Goal Clarification	H_0 (Null Hypothesis): Responses are equally distributed across all categories	Chi-square test	**Null not rejected** - Equal distribution across all categories
Q9: Job Market Confidence	H0: Students feel moderately confident about job readiness (rating ≥ 3)	One-sample t-test	**Null not rejected** - Students are confident about job interviews or job market readiness after the internship,

Q11: Pre-placement offer (PPO) from internship organization	H_0 (Null Hypothesis): P_intern = P_no_intern (The probability of securing a future job is the same whether or not you do an internship.)	Chi-square test	**Null not rejected** - No statistically significant evidence that the probability of securing a future job differs between interns and non-interns based on this sample.
Q13: Leverage internship connections for future career opportunities	H_0 (Null Hypothesis): Students are likely to leverage internship connections for future career opportunities (rating >= 3)	One-sample t-test	**Null not rejected** - Students are likely to leverage internship connections for future career opportunities
Q14: Understanding of Industry Expectations	H_0 (Null Hypothesis): Internship helps to understand industry expectations and professional standards (rating ≥ 3)	One-sample t-test	**Null not rejected** - internships help students understand industry expectations and professional standards
Q16: Overall Internship satisfaction	H0 (Null Hypothesis): Students are satisfied with their internship	One-sample t-test	**Null not rejected** - Students are satisfied their internship

	experience in terms of career development (rating >= 3)		

8. Conclusion

This study provides compelling evidence that IIT internships have a significant positive impact on career development. The findings demonstrate that the internship program effectively bridges the academic-industry gap, with students reporting high levels of skill acquisition, career clarity, and job market readiness.

Key findings include:

1. **Skill Development**: The majority of students acquired valuable technical skills, particularly in software development tools and frameworks, with strong improvements in soft skills like teamwork and communication.
2. **Career Clarity**: Internships significantly help students clarify their career goals and explore different career paths, with many discovering new interests or confirming existing ones.
3. **Job Market Readiness**: Students show significantly higher confidence in job interviews and market readiness after completing their internships, with substantial improvements in their resumes and portfolios.
4. **Professional Networking**: Students successfully build valuable professional connections, with 62.5% receiving pre-placement offers, indicating strong industry engagement.
5. **Industry Understanding**: Internships are highly effective in helping students understand industry expectations and professional standards.

The statistical analysis confirms that IIT internships exceed moderate effectiveness across most career development dimensions. While students show moderate likelihood of leveraging networking connections, the overall impact on career development is substantial and positive.

These findings suggest that the IIT internship program successfully fulfills its purpose of preparing students for professional careers and should be continued with potential enhancements in networking guidance and long-term connection maintenance.

9. References

Survey questionnaire data: Software Metrics Assignment - Form Responses (Internal data)

Statistical analysis documentation: [Available upon request]

Response timeline: June 2, 2025 - July 7, 2025

Total responses: 17 students from various BSSE batches

Google Sheet: Software Metrics Assignment

Google Form: https://forms.gle/2mQzSYX7bpne46NF8